



SEMESTER END EXAMINATIONS - SEPTEMBER / OCTOBER 2023

Program : **B.E :- Common to ECE / EEE / EIE / ETE / MLE**
Course Name : **Differential Equations and Laplace**
Course Code : **MAE21**

Semester : **II**
Max. Marks : **100**
Duration : **3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit

UNIT - I

1. a) Define maxima and minima for the function of two independent variables. CO1 (02)
b) Determine the extreme value of $f(x, y) = xy(1 - x - y)$ at $(1/3, 1/3)$. CO1 (04)
c) Find the Maclaurin's series for $\tan x$ upto the term containing x^4 . CO1 (07)
d) Compute the real root of the equation $x \log_{10} x = 1.2$, by Newton-Raphson method, correct to three decimal places. CO1 (07)

2. a) Write the formula to find the Taylors series expansion of functions of two variables. CO1 (02)
b) Solve the system of non linear equations $y^2 + x = 7, x^2 + y = 11$, at $(3.5, -1.8)$ by Newton-Raphson method. Carryout one iteration. CO1 (04)
c) Obtain the Taylor's series expansion of $xy + \sin(xy)$ about $(1, \frac{\pi}{2})$ upto the second degree terms. CO1 (07)
d) Find the extreme value of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$. CO1 (07)

UNIT - II

3. a) Define orthogonal trajectory of family of curves. CO2 (02)
b) A bungee jumper with a mass of 68.1 Kgs leaps from a stationary hot air balloon. Use $\frac{dv}{dt} = g - \frac{cv^2}{m}$ where $g = 9.8m/s^2, c = 0.25kg/m$ to compute velocity for the third second of free fall by Euler's method in steps of 1 second. CO2 (04)
c) Solve $(y^3 - 3xy^2)dx + (2x^2y - xy^2)dy = 0$. CO2 (07)
d) An emf $e = 200e^{-5t}$ is applied to a series circuit consisting of 20 ohm resistor and 0.01 F capacitor. Find the charge and current at any time assuming that there is no initial charge on capacitor. CO2 (07)

4. a) Write the steps involved in finding the orthogonal trajectories of the family of curves in Polar form. CO2 (02)
b) Solve $(1 + 2xy \cos x^2 - 2xy)dx + (\sin x^2 - x^2)dy = 0$. CO2 (04)
c) Show that the family of curves $\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1$ is self-orthogonal, where λ is the parameter. CO2 (07)
d) Solve initial value problem $\frac{dy}{dx} = xy + y^2; y(0) = 1$ at $x = 0.4$ taking $h=0.4$ using Runge-Kutta method of fourth order. CO2 (07)

UNIT - III

- ✓ 5. a) Define a boundary value problem with an example. CO3 (02)
 b) Solve the linear differential equation $(2x+3)^2 y'' + 2(2x+3)y' - 12y = 0$. CO3 (04)
 c) Solve the equation $(D^3 - 7D - 6)y = e^{2x}(1+x)$, where $D = d/dx$. CO3 (07)
 d) Solve the differential equation $y'' - 2y' + 2y = e^x \tan x$ by the method of variation of parameters. CO3 (07)
6. a) Define Linear and nonlinear differential equation. CO3 (02)
 b) If $D = \frac{d}{dx}$ and $X = X(x)$, then prove that $\frac{1}{D+a} X = e^{-ax} \int X e^{ax} dx$. CO3 (04)
 c) Solve the equation $(D^2 - 4D + 3)y = \sin 3x \cdot \cos 2x$, where $D = d/dx$ CO3 (07)
 d) Solve the Cauchy's linear differential equation
 $x^2 y'' - 4xy' + 6y = \cos[2 \log x]$ CO3 (07)

UNIT- IV

7. a) Define beta function and hence write the trigonometric form of beta function. CO4 (02)
 b) Evaluate $\int_0^{\infty} e^{-2t} t^2 \sin t \, dt$ using Laplace transform. CO4 (04)
 c) Find (i) $L\left\{\frac{\sin t \sin 3t}{t}\right\}$ (ii) $e^{-3t} \cos 5t \sin 5t$. CO4 (07)
 d) If $f(t)$ and its $(n-1)$ derivatives are continuous then prove that
 $L\{f^{(n)}(t)\} = s^n L\{f(t)\} - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0)$. CO4 (07)
- ✓ 8. a) Evaluate $\beta\left(\frac{1}{3}, \frac{2}{3}\right)$. CO4 (02)
 b) Find the Laplace transform of $(t^2 + 4)^2 + \sinh 2t \sin 4t$. CO4 (04)
 c) Show that $\int_0^{\infty} \frac{e^{-x^2}}{\sqrt{x}} dx \times \int_0^{\infty} e^{-x^2} x^2 dx = \frac{\pi}{4\sqrt{2}}$. CO4 (07)
 d) Find the Laplace transform of : CO4 (07)
 i) $\left(\sqrt{t} + \frac{1}{\sqrt{t}}\right)^3$ ii) $\int_0^t t e^{-t} \cos t \, dt$.

UNIT- V

- ✓ 9. a) Define dirac delta function. CO5 (02)
 b) Find the inverse Laplace transform of $\frac{1}{(s+1)(s^2+4)}$. CO5 (04)
 c) Find the Laplace transform of the triangular wave of period $2a$ given by
 $f(t) = \begin{cases} t, & 0 < t < a \\ 2a-t, & a < t < 2a \end{cases}$ CO5 (07)
 d) Using Laplace transform techniques, solve
 $y' - 3x = e^{3t}, \quad x' - 3y = e^{-t}, y(0) = 0, x(0) = 0$ CO5 (07)

10. a) State convolution theorem. CO5 (02)
b) Find the inverse Laplace transform of $\frac{s}{s^2+6s+13}$. CO5 (04)
c) Express the following function in terms of unit step function and hence find its Laplace transform of $f(t) = \begin{cases} 1, & 0 < t < 1 \\ 2t, & 1 < t < 2 \\ 3t, & 2 < t < 3 \end{cases}$ CO5 (07)
d) Using Laplace transform techniques, solve $y'' - 3y' + 2y = e^{3t}$, $y(0) = 1$, $y'(0) = -1$. CO5 (07)
